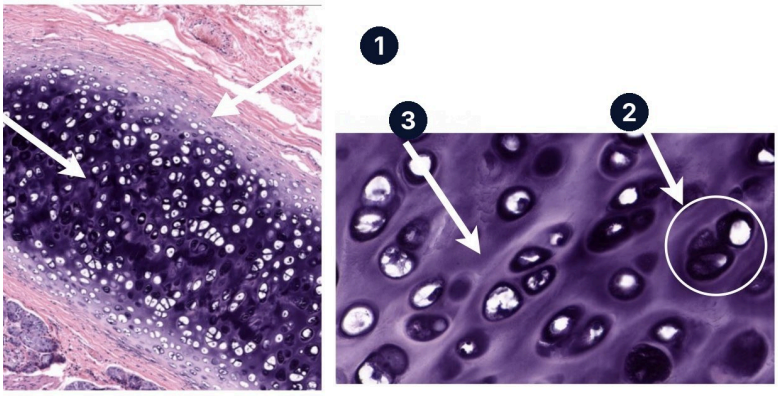
BIO 004 ANATOMY ATLAS · CARTILAGE, BONE TISSUE AND THE LONG BONE · CARD 1 · CARTILAGE AND BONE TISSUE

PRACTICE

Hyaline cartilage



1. Name this structure.

2. Name this structure.

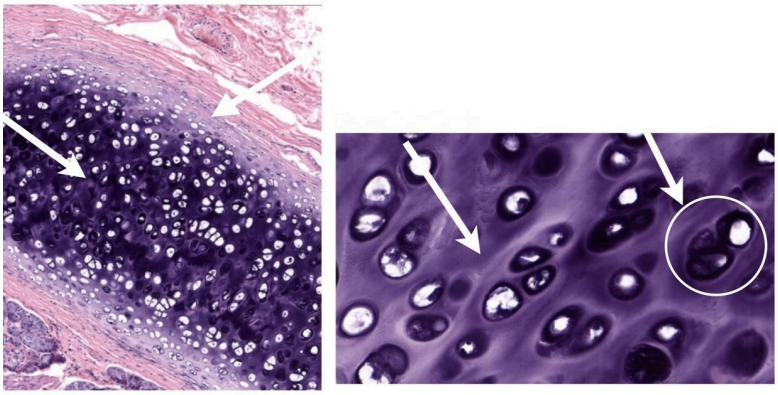
3. Name this structure.

Write each answer on its line with a dry-erase marker, then flip the card to check.

BIO 004 ANATOMY ATLAS · CARTILAGE, BONE TISSUE AND THE LONG BONE · CARD 2 · CARTILAGE AND BONE TISSUE

PRACTICE

Name this tissue



1. Name this tissue.

2. What does it look like?

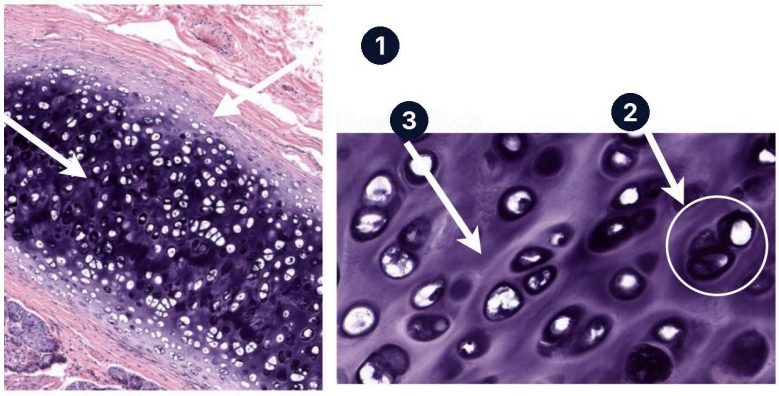
3. Where is it found?

4. What does it do?

Write each answer on its line with a dry-erase marker, then flip the card to check.

ANSWER KEY

Hyaline cartilage

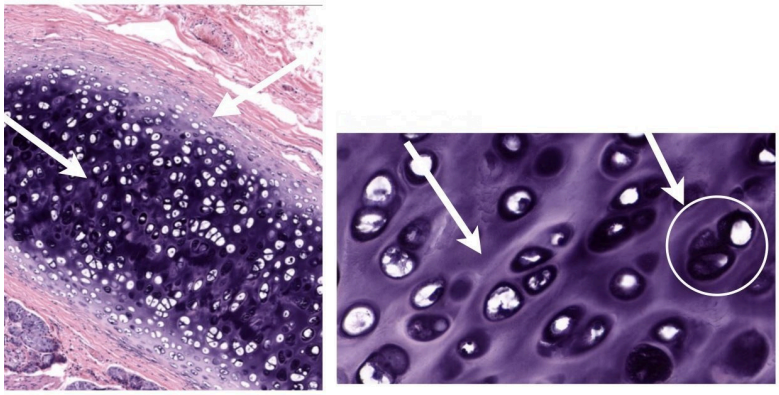


1. Name this structure.
   Perichondrium
2. Name this structure.
   Lacuna with chondrocyte
3. Name this structure.
   Extracellular matrix

Check your answers, then wipe the front clean for the next round.

ANSWER KEY

Name this tissue



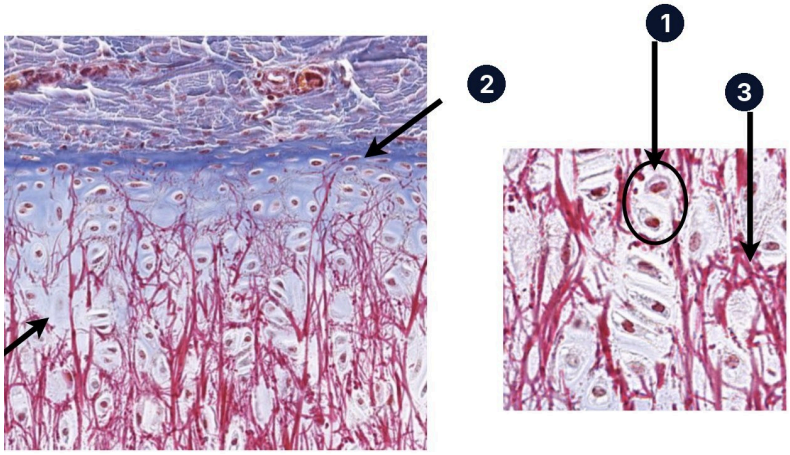
1. Name this tissue.
   Hyaline cartilage
2. What does it look like?
   Glassy, smooth matrix with chondrocytes in lacunae; perichondrium around it
3. Where is it found?
   Nose, trachea, larynx, ends of long bones (articular cartilage), fetal skeleton
4. What does it do?
   Smooth joint surfaces, flexible support; most common cartilage

Check your answers, then wipe the front clean for the next round.

BIO 004 ANATOMY ATLAS · CARTILAGE, BONE TISSUE AND THE LONG BONE · CARD 3 · CARTILAGE AND BONE TISSUE

PRACTICE

Elastic cartilage



1. Name this structure.

2. Name this structure.

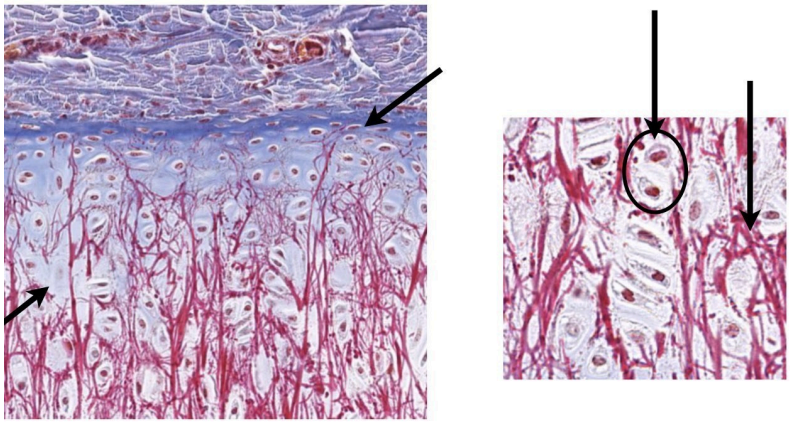
3. Name this structure.

Write each answer on its line with a dry-erase marker, then flip the card to check.

BIO 004 ANATOMY ATLAS · CARTILAGE, BONE TISSUE AND THE LONG BONE · CARD 4 · CARTILAGE AND BONE TISSUE

PRACTICE

Name this tissue



1. Name this tissue.

2. What does it look like?

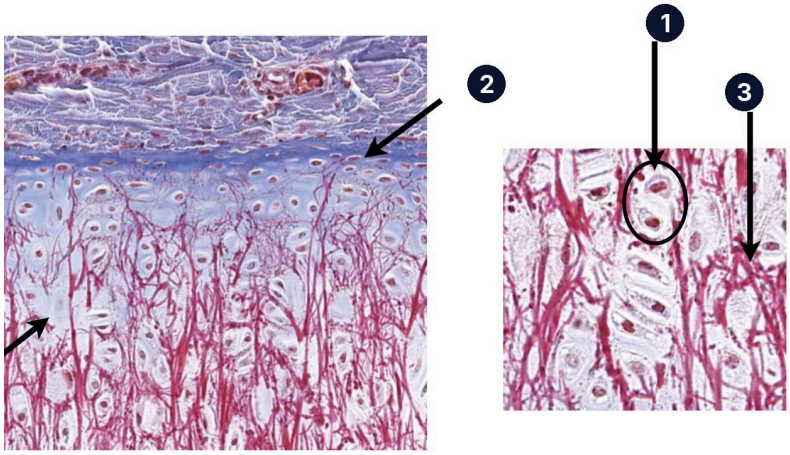
3. Where is it found?

4. What does it do?

Write each answer on its line with a dry-erase marker, then flip the card to check.

ANSWER KEY

Elastic cartilage

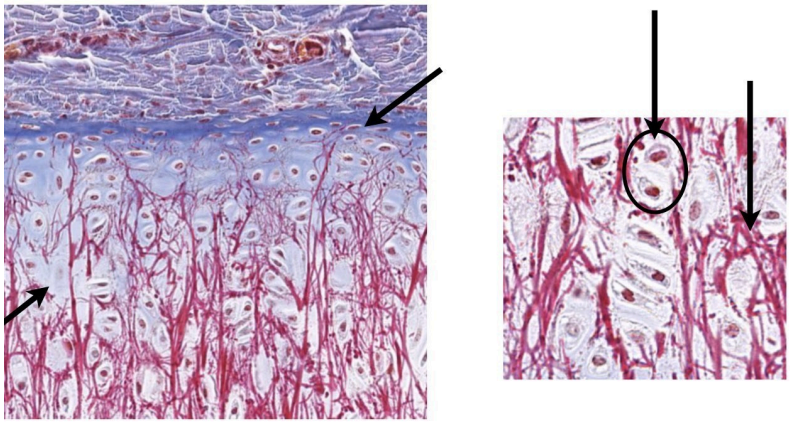


1. Name this structure.
   Lacuna with chondrocyte
2. Name this structure.
   Perichondrium
3. Name this structure.
   Elastic fibers

Check your answers, then wipe the front clean for the next round.

ANSWER KEY

Name this tissue



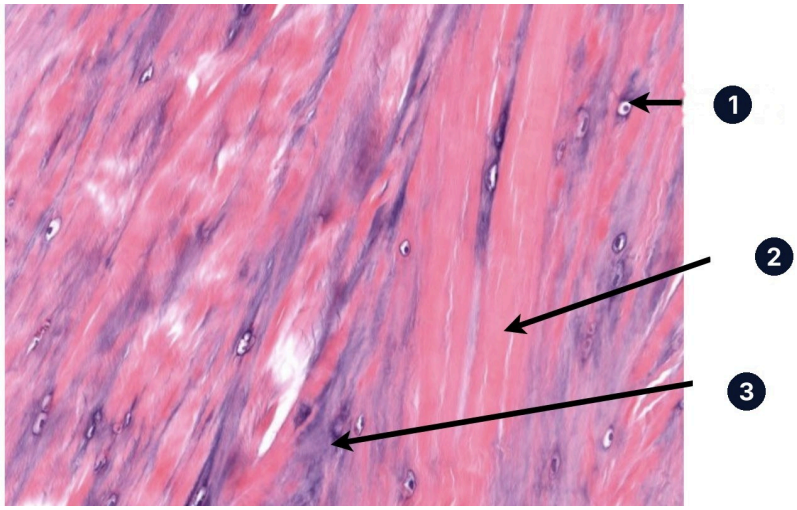
1. Name this tissue.
   Elastic cartilage
2. What does it look like?
   Chondrocytes in lacunae surrounded by many dark elastic fibers
3. Where is it found?
   External ear (auricle) and epiglottis
4. What does it do?
   Gives strength and flexibility; keeps its shape after bending

Check your answers, then wipe the front clean for the next round.

BIO 004 ANATOMY ATLAS · CARTILAGE, BONE TISSUE AND THE LONG BONE · CARD 5 · CARTILAGE AND BONE TISSUE

PRACTICE

Fibrocartilage



1. Name this structure.

2. Name this structure.

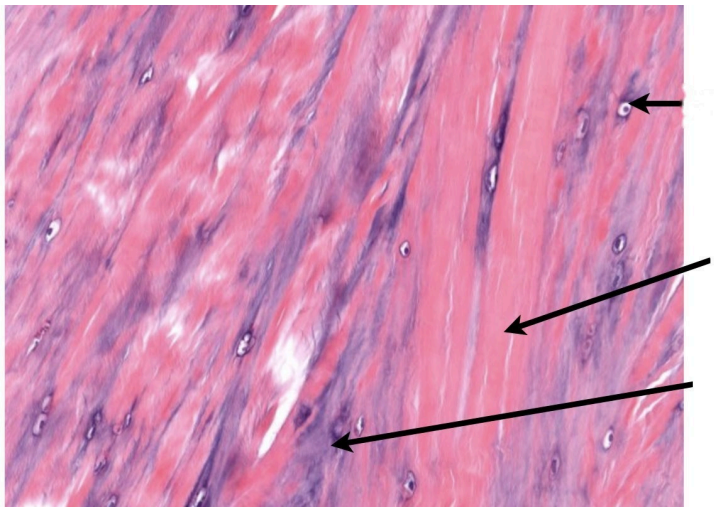
3. Name this structure.

Write each answer on its line with a dry-erase marker, then flip the card to check.

BIO 004 ANATOMY ATLAS · CARTILAGE, BONE TISSUE AND THE LONG BONE · CARD 6 · CARTILAGE AND BONE TISSUE

PRACTICE

Name this tissue



1. Name this tissue.

2. What does it look like?

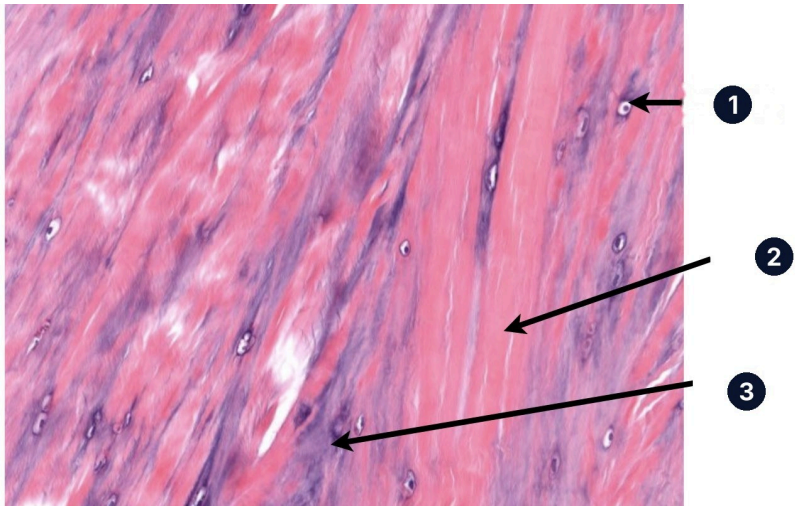
3. Where is it found?

4. What does it do?

Write each answer on its line with a dry-erase marker, then flip the card to check.

ANSWER KEY

Fibrocartilage

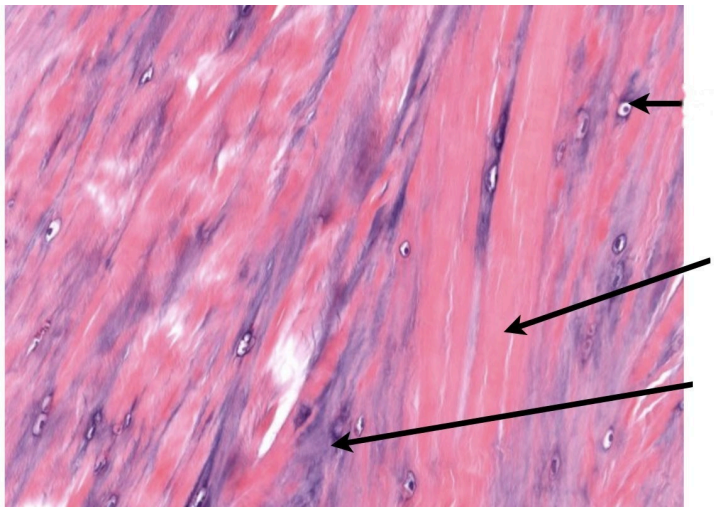


1. Name this structure.
   Lacuna with chondrocyte
2. Name this structure.
   Collagen fibers
3. Name this structure.
   Extracellular matrix

Check your answers, then wipe the front clean for the next round.

ANSWER KEY

Name this tissue

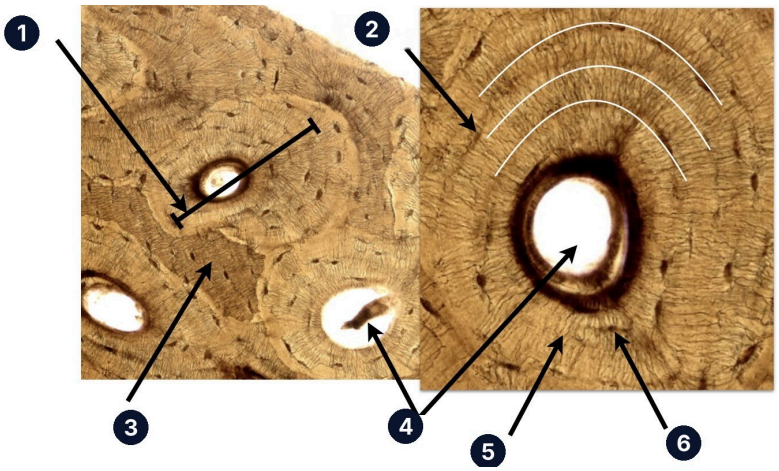


1. Name this tissue.
   Fibrocartilage
2. What does it look like?
   Rows of chondrocytes between thick parallel collagen fibers; no perichondrium
3. Where is it found?
   Intervertebral discs, pubic symphysis, menisci of the knee, TMJ
4. What does it do?
   Absorbs shock and resists compression

Check your answers, then wipe the front clean for the next round.

PRACTICE

Compact bone (osseous tissue)

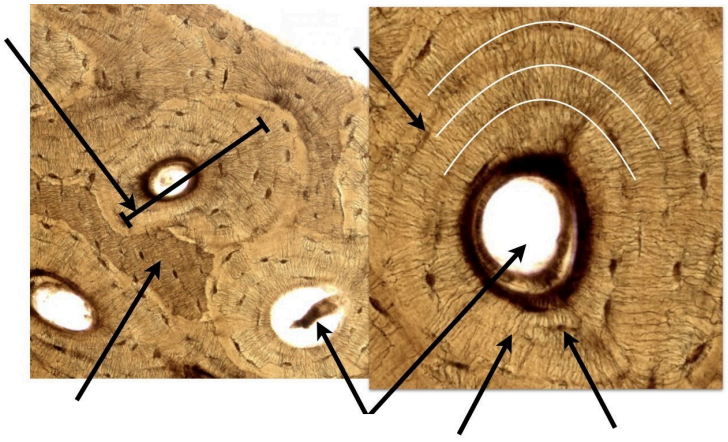


1. Name this structure.
2. Name this structure.
3. Name this structure.
4. Name this structure.
5. Name this structure.
6. Name this structure.

Write each answer on its line with a dry-erase marker, then flip the card to check.

PRACTICE

Name this tissue

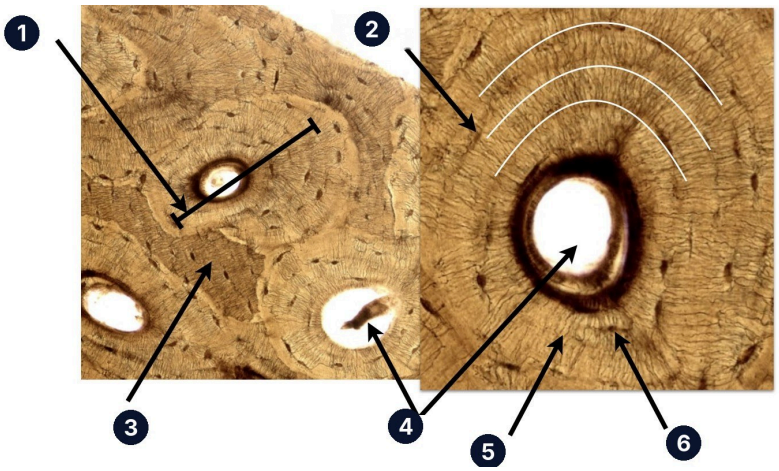


1. Name this tissue.
2. What does it look like?
3. Where is it found?
4. What does it do?

Write each answer on its line with a dry-erase marker, then flip the card to check.

ANSWER KEY

Compact bone (osseous tissue)

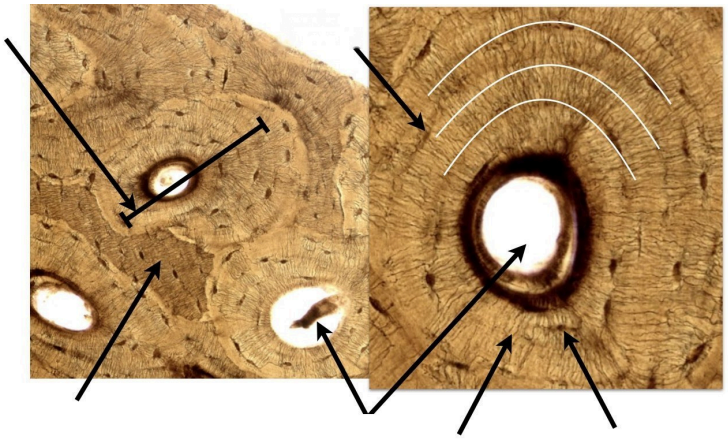


1. Name this structure.
   Osteon
2. Name this structure.
   Concentric lamellae
3. Name this structure.
   Interstitial lamellae
4. Name this structure.
   Central (Haversian) canal
5. Name this structure.
   Canaliculi
6. Name this structure.
   Lacuna

Check your answers, then wipe the front clean for the next round.

ANSWER KEY

Name this tissue



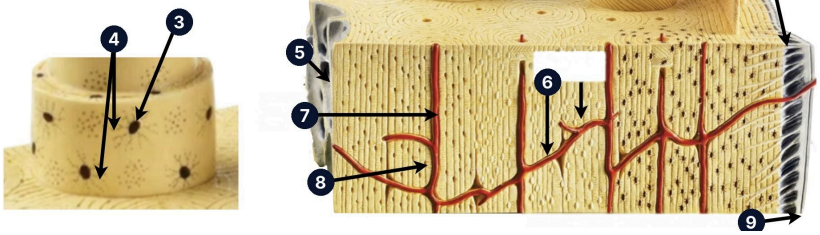
1. Name this tissue.
   Compact bone (osseous tissue)
2. What does it look like?
   Osteons: rings of lamellae around a central canal, with osteocytes in lacunae linked by canaliculi
3. Where is it found?
   Diaphysis (shaft) of long bones and the outer layer of all bones
4. What does it do?
   Support and protection, movement (lever for muscles), mineral storage

Check your answers, then wipe the front clean for the next round.

BIO 004 ANATOMY ATLAS · CARTILAGE, BONE TISSUE AND THE LONG BONE · CARD 9 · CARTILAGE AND BONE TISSUE

PRACTICE

Microanatomy of compact bone (model)



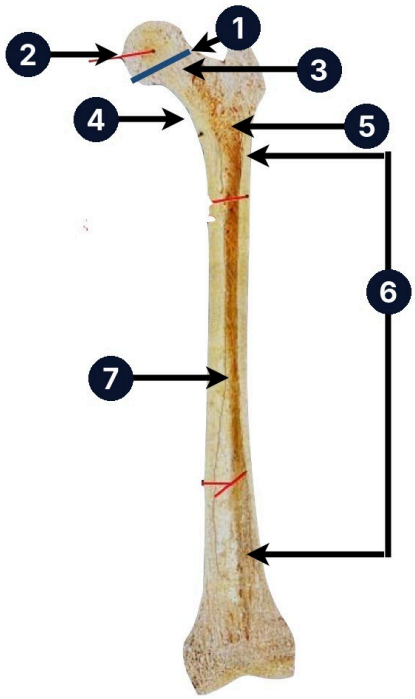
1. Name this structure.
2. Name this structure.
3. Name this structure.
4. Name this structure.
5. Name this structure.
6. Name this structure.
7. Name this structure.
8. Name this structure.
9. Name this structure.

Write each answer on its line with a dry-erase marker, then flip the card to check.

BIO 004 ANATOMY ATLAS · CARTILAGE, BONE TISSUE AND THE LONG BONE · CARD 10 · LONG BONE

PRACTICE

Parts of a long bone



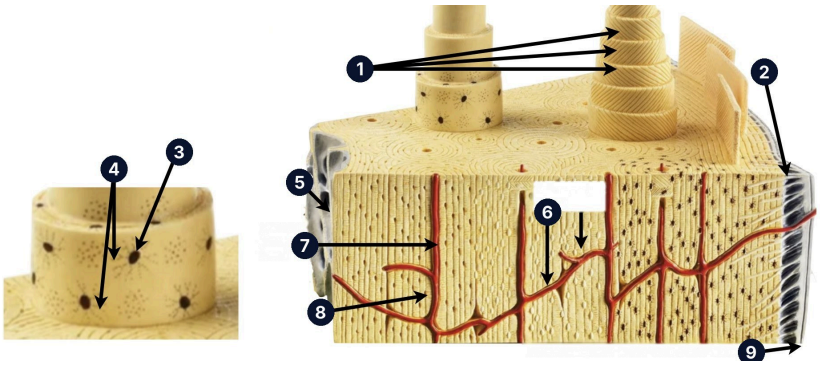
1. Name this part of the long bone.
2. Name this part of the long bone.
3. Name this part of the long bone.
4. Name this part of the long bone.
5. Name this part of the long bone.
6. Name this part of the long bone.
7. Name this part of the long bone.

Write each answer on its line with a dry-erase marker, then flip the card to check.

BIO 004 ANATOMY ATLAS · CARTILAGE, BONE TISSUE AND THE LONG BONE · CARD 9 · CARTILAGE AND BONE TISSUE

ANSWER KEY

Microanatomy of compact bone (model)



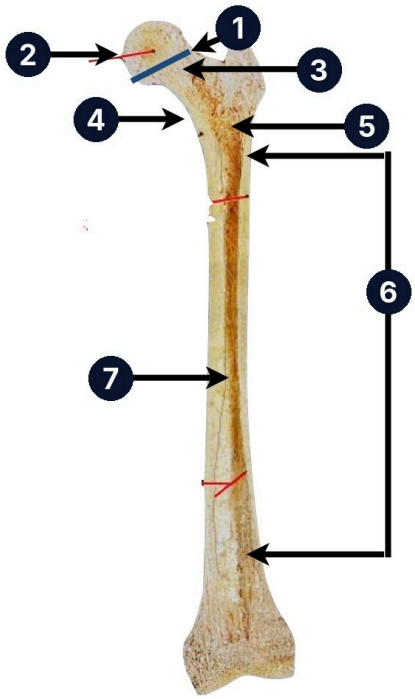
1. Name this structure.
   Concentric lamellae
2. Name this structure.
   Sharpey's fibers
3. Name this structure.
   Lacuna with osteocyte
4. Name this structure.
   Canaliculi
5. Name this structure.
   Spongy bone
6. Name this structure.
   Perforating (Volkmann) canal
7. Name this structure.
   Central (Haversian) canal
8. Name this structure.
   Lacuna (empty)
9. Name this structure.
   Periosteum

Check your answers, then wipe the front clean for the next round.

BIO 004 ANATOMY ATLAS · CARTILAGE, BONE TISSUE AND THE LONG BONE · CARD 10 · LONG BONE

ANSWER KEY

Parts of a long bone

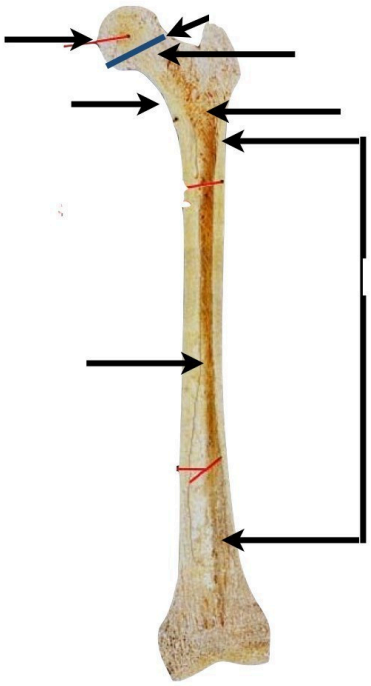


1. Name this part of the long bone.
   Epiphyseal line
2. Name this part of the long bone.
   Articular cartilage
3. Name this part of the long bone.
   Epiphysis
4. Name this part of the long bone.
   Periosteum
5. Name this part of the long bone.
   Metaphysis
6. Name this part of the long bone.
   Diaphysis
7. Name this part of the long bone.
   Medullary cavity

Check your answers, then wipe the front clean for the next round.

PRACTICE

Long bone: quick check



1. What fills the medullary cavity in adults?

2. What fills the epiphysis, and what process happens there?

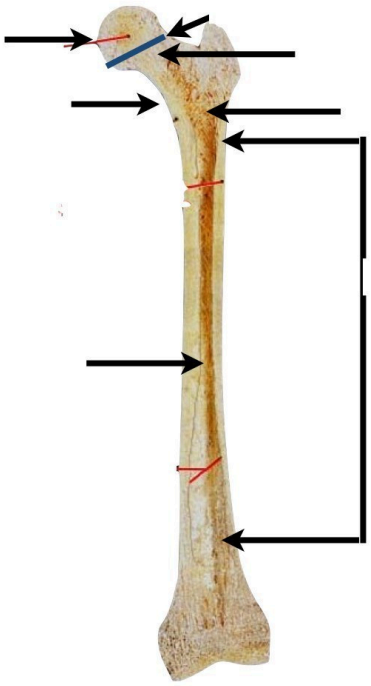
3. Epiphyseal plate vs. epiphyseal line?

4. What are the two layers of the periosteum?

Write each answer on its line with a dry-erase marker, then flip the card to check.

ANSWER KEY

Long bone: quick check



1. What fills the medullary cavity in adults?

   Yellow (fatty) marrow

2. What fills the epiphysis, and what process happens there?

   Red marrow in spongy bone; hematopoiesis (blood cell formation)

3. Epiphyseal plate vs. epiphyseal line?

   Plate: hyaline cartilage growth zone in children. Line: bony remnant in adults

4. What are the two layers of the periosteum?

   Outer fibrous layer and inner osteogenic layer

Check your answers, then wipe the front clean for the next round.